ENEL 585 – Lab 2

DC-AC Conversion: Single-Phase Full-Bridge Inverter — Square-Wave, Phase-Shift and Sinusoidal PWM Modulation

Group Information

Group #: _____

Student name(s): _____

Purpose

To operate both legs of the inverter box as a single-phase full-bridge inverter that supplies a series RL load (R = 12.5 Ω, L = 5 mH) from a 55 V DC link, and to compare four modulation schemes at a 60 Hz fundamental: square-wave, phase-shift, bipolar sinusoidal PWM and unipolar sinusoidal PWM. For each scheme you will determine how the modulation command (α or m_i) sets the fundamental component of the output voltage, identify the frequencies of the dominant harmonics, and compare the measured harmonic currents with those predicted by the Fourier analysis from class. You will also measure the power factor at the bridge output, and compare harmonic measurements from the Fluke 39 power meter with those from the oscilloscope FFT to establish the frequency range over which each instrument is valid.

Marking

Total Marks for Pre-Lab Quiz = __ /10 (Weight = 2%)
Total Marks for Lab Report = __ /10 (Weight = 2%)
Total Marks for Lab Work = __ /10 (Weight = 2%)

The pre-lab itself is not collected or graded. Instead, a **10-minute individual quiz** is written at the start of the lab session, based directly on the pre-lab questions below.

Associated Documents

Lab 2 uses the same inverter box, interface board, LaunchPad and bench instruments as Lab 1. This handout assumes you can wire the control side, build and flash a program and use External Mode without step-by-step instructions; the guides below carry the detail. What is new is the second inverter leg, the RL load between the two leg midpoints, and the Fluke 39 power meter.

- *PLECS Setup* — building and flashing the LaunchPad, using External Mode.
- *Inverter Box Setup Guide* — wiring the inverter box, interface board and LaunchPad: auxiliary +15 V power, thermal trip and gate-drive BNCs.
- *Bench Instruments Guide* — front-panel procedures for the E3631A, the ITECH IT-M3632, the oscilloscope and probes (including the FFT), and the Fluke 39.
- PLECS simulation models (on D2L), one per scheme: square_wave_mod.plecs, phase_shift_mod.plecs, SPWM_bipolar.plecs and SPWM_unipolar.plecs. Use them to check your pre-lab answers.
- LaunchPad programs (on D2L): square_phase_shift_ti_launchpad.plecs (square wave and phase shift, Part A), spwm_bipolar_ti_launchpad.plecs and spwm_unipolar_ti_launchpad.plecs (Part B).

Introduction

Fig. 1 shows the circuit. Each leg of the inverter box is a half-bridge like the one you used in Lab 1: leg A is TOP1 and BOT1 (S_1 and S_3 in the lecture notation), leg B is TOP2 and BOT2 (S_2 and S_4). The load — the 5 mH inductor in series with the resistor load bank — is connected between the two leg midpoints A and B, so the bridge output voltage is $v_t = v_{AN} - v_{BN}$. The ITECH supply holds the DC link at v_d = 55 V and delivers the average power. The bridge draws a pulsating current, and during parts of each cycle the inductive load returns energy to the DC link: the 1.1 mF DC-link capacitor inside the inverter box supplies and absorbs these exchanges.

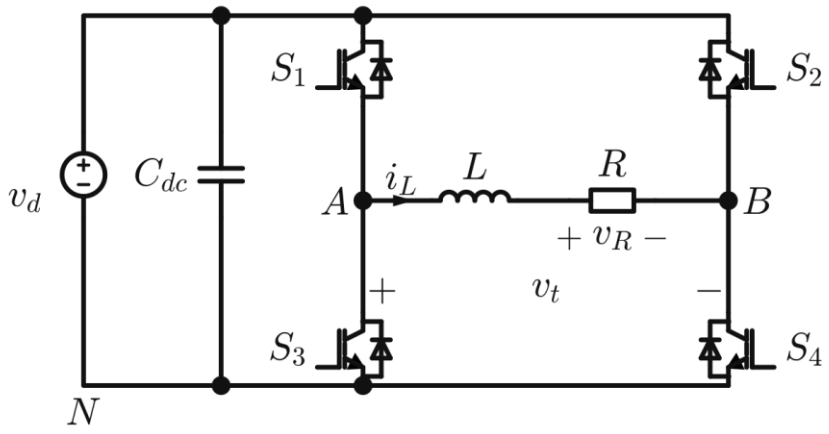


Fig. 1 – Single-phase full bridge formed from legs A and B of the inverter box (S_1 = TOP1, S_3 = BOT1, S_2 = TOP2, S_4 = BOT2), feeding a series RL load. v_t is measured from A to B; N is the DC− rail.

The two devices in each leg are always gated complementarily, with a 5 µs dead time (both off) inserted by the LaunchPad at every transition. The four schemes differ only in the gate pattern:

- **Square wave:** each device is gated on for half of the fundamental period T_1, and the two legs are half a period apart, so $v_t = \pm v_d$. There is no amplitude command: the fundamental is fixed by v_d. It is the phase-shift pattern with $\alpha = 0$, so one program runs both.
- **Phase shift:** the diagonal gate signals S_1 and S_4 are shifted in opposite directions, so v_t is zero for an interval αT_1 in each half period (lecture notation). $\alpha = 0$ is the square wave; $\alpha = 0.5$ gives zero output.
- **Bipolar SPWM:** a 60 Hz sinusoidal reference of amplitude m_i is compared with a 4.98 kHz triangular carrier (83 carrier periods per 60 Hz period). The diagonal pairs (S_1, S_4) and (S_2, S_3) switch together, so $v_t = \pm v_d$.
- **Unipolar SPWM:** each leg compares its own reference with the same carrier (the sinusoid for leg A, its negative for leg B), so v_t takes the levels $+v_d$, 0 and $-v_d$.

The programs are open loop: the modulation command (α or m_i) is a tunable parameter in External Mode, and nothing is measured by the LaunchPad. There is **no firmware voltage or current trip** in this lab. Protection comes from the ITECH's current limit and over-voltage protection, the inverter box's thermal trip and its gate drivers' built-in protection — which is why the settings below matter.

Pre-Lab

Work these before the session; the pre-lab quiz at the start of the lab is drawn from them. Keep your calculated values: your report compares the measurements with them. Use the harmonic results and the SPWM harmonic table from the inverter lecture.

Parameters (all questions): v_d = 55 V, R = 12.5 Ω, L = 5 mH (series RL load, Fig. 1), f_1 = 60 Hz, f_{sw} = 4.98 kHz (83f_1) for SPWM. Assume ideal switches and zero dead time. Give harmonic voltages and currents as rms values. $I_{L,h}$ is the rms value of the h-th harmonic of i_L (h = 1 is the fundamental), and the **fundamental power** is $P_1 = I_{L,1}^2 R$, the power delivered to R by the fundamental alone. "The harmonic with the largest amplitude" excludes the fundamental.

1. **Square wave.** (a) Sketch v_t and i_L over one fundamental period, labelling the voltage levels and the switching instants, and justify the shape of i_L. (b) Calculate $I_{L,1}$, P_1, the 3rd-harmonic voltage $V_{t,3}$ and $I_{L,3}$.
2. **Phase shift.** Find the α at which the fundamental of v_t has a peak of 0.88v_d (the fundamental that sinusoidal PWM produces at m_i = 0.88, used in Part B), and calculate $I_{L,1}$, P_1 and $I_{L,3}$ at that α.
3. **Bipolar SPWM at m_i = 0.88.** (a) Find the harmonic of v_t with the largest amplitude: its order, frequency and voltage, and the current it drives through the load. Compare its voltage and current with the 3rd harmonic in question 1 and explain the difference. (b) In bipolar SPWM $|v_t| = v_d$ at every instant, so $V_{t,rms} = v_d$. Neglect the harmonic currents ($I_{L,h} = 0$ for $h > 1$, so $I_{L,rms} = I_{L,1}$ and $P = P_1$) and calculate the power factor at the bridge output, $P/(V_{t,rms} I_{L,rms})$. Compare it with the load's displacement factor cos ϕ_1.
4. **Unipolar SPWM at m_i = 0.88.** Which groups of bipolar-SPWM harmonics are absent from the unipolar v_t? At what frequencies are the harmonics of v_t with the largest amplitude?

Safety Notes

- The DC link stores energy after shutdown: 1.1 mF at 55 V. Ramping the ITECH voltage to zero before switching its output off discharges it through the source. Before touching or re-wiring the circuit, confirm the DC link is below 10 V. The ITECH is wired directly across the link, so its voltage reading is the link voltage: read it during the ramp, before turning the output off. If the ITECH leads have been disconnected, its reading no longer shows the link — measure across the DC-link terminals instead.
- Make every wiring change — including moving a probe — with switching disabled and the ITECH output off. Never re-wire an energized circuit.
- Neither load terminal is at DC− potential: both leg midpoints switch between the DC rails. Measure v_t and v_R only with the differential probes.
- Set the ITECH limits before connecting it: source mode, 55 V, 10 A current limit, OVP 60 V. The ITECH is connected directly across the DC link. If the link voltage ever exceeds 60 V, the OVP turns the ITECH output off — it cannot pull the link down, so the link then discharges only through the internal balancing resistors (time constant about 46 s). Do not raise v_d above 55 V.
- Switching is controlled from enableswitch in External Mode, which selects the disable (0) or enable (1) constant — do not edit the constants. Select disable before the ITECH output is turned on, enter the starting command (α or m_i), and enable only once v_d is at 55 V. Start Part A at α = 0.5 (zero output) and Part B at m_i = 0.2, then move to the test value; α = 0 applies the full square wave.
- Keep the load connected while switching. Change the load bank one switch at a time with the others closed, never open every switch, and never disconnect the inductor while current flows. Obtain TA sign-off before applying power.

Equipment

Component	Purpose	Qty
Oscilloscope (Keysight DSOX2004A)	Display probed signals; FFT of i_L	1
Keysight 1146B clamp-on current probe	Measure i_L	1
Keysight N2791A differential voltage probe	Measure v_t and v_R	2
BNC splitter	Share the current-probe signal between the oscilloscope and the power meter	1
Fluke 39 power meter	Harmonics, THD and power at the 60 Hz fundamental	1
Inverter box (SEMITEACH)	Legs A (TOP1/BOT1) and B (TOP2/BOT2) and the internal 1.1 mF DC-link capacitor	1
Interface board + LAUNCHXL-F28P55X	Gate drive and modulation (see Inverter Box Setup Guide)	1
Laptop with PLECS + TI C2000 TSP	Build, flash and adjust the modulator (see PLECS Setup)	1
5 mH inductor	L in Fig. 1	1
Resistor load bank	R in Fig. 1: 20 × 250 Ω in parallel, one switch each — 1 switch = 250 Ω, 5 = 50 Ω, 10 = 25 Ω, all 20 = 12.5 Ω	1
Keysight/Agilent E3631A supply	+15 V auxiliary power for gate drivers and interface board	1
ITECH IT-M3632	DC-link source v_d (55 V)	1
Banana and BNC cables	Wiring per the setup guides	—

Instrument Settings for Lab 2

Front-panel procedures are in the *Bench Instruments Guide*. Set every limit before the instrument is wired into the circuit, with its output off. The TA checks these settings at sign-off:

- **E3631A:** +25V output at 15.00 V with a 1.000 A limit (Guide §1).
- **ITECH IT-M3632:** source mode, 55 V, 10 A current limit, OVP 60 V, output off until startup (Guide §3).
- **Oscilloscope:** set it up by hand — do not use [Auto Scale] — and read rms and average values with [Meas] over whole fundamental periods: use the Full Screen variants with a whole number of periods on screen, e.g. three at 5 ms/div (Guide §4). Voltage probes 10:1 (±70 V range, enough for v_t: ±55 V plus switching overshoot); current probe 100 mV/A. Channel 1 v_t, channel 2 v_R, channel 3 i_L. Starting settings are in Table 1.
- **Oscilloscope FFT** ([Math] → FFT): source i_L, Flat Top window (it gives the most accurate peak amplitudes), 20 ms/div so that the record holds twelve fundamental periods, span 0 to 12.5 kHz, 10 dB/div. Read each peak with the cursors. A level of L dB converts to an rms value of $10^{L/20}$ in the unit shown (V, or A when the channel units are set to Amps). Procedure and pitfalls: Guide §4, FFT.

- **Fluke 39** (Guide §5): voltage leads across R (red on the inductor side); current input from the BNC splitter. At power-on, open the setup screen and check THD = %F (THD relative to the fundamental; the factory setting is %RMS) and CLAMP = OTHER (flat response for a non-Fluke clamp). The meter expects a 1 mV/A clamp, so with the 1146B at 100 mV/A **it displays current and power 100 times too large** — divide by 100. It analyses harmonics up to the 31st (1.86 kHz at 60 Hz), and its input bandwidth is about 2.1 kHz.

Table 1: Suggested starting oscilloscope settings — adjust as needed

View	Time/div	v_t and v_R	i_L	Trigger
Exp. 1–2: square wave, phase shift	5 ms	20 V/div, 0 V at centre	2 A/div, 0 A at centre	v_t, rising, 0 V, Auto
Exp. 3–5: SPWM, fundamental	5 ms	20 V/div, 0 V at centre	2 A/div, 0 A at centre	v_R, rising, 0 V, Auto
Exp. 3–5: SPWM, switching detail	100 µs	20 V/div, 0 V at centre	DC coupling, 500 mA/div, offset to a current peak	v_t, rising, Auto; press [Stop] to freeze
FFT of i_L	20 ms	—	FFT: 0–12.5 kHz, 10 dB/div	Auto

Setup — Full-Bridge Inverter

The control side is wired as in Lab 1; the new parts are the second leg's gate BNCs and the load between the leg midpoints. The TA checks the wiring before power is applied.

Hardware (Fig. 1)

1. Control side, per the *Inverter Box Setup Guide*: seat the LaunchPad on the interface board, connect the +15 V auxiliary supply and wire the thermal-trip loop and the ERROR1 and ERROR2 BNCs as in Lab 1. Connect **all four gate BNCs** of inverter 1 — Top 1, Bot 1, Top 2 and Bot 2 (labels match on both ends).
2. Power circuit of Fig. 1 with banana cables: ITECH output across the inverter DC-link terminals (mind polarity), then leg-A midpoint → 5 mH inductor → resistor load bank → leg-B midpoint. Close **all 20 load-bank switches** (12.5 Ω). Nothing else connects to the DC-link terminals.
3. Configure the ITECH (source mode, 55 V, 10 A, OVP 60 V) and leave its output off.
4. Voltage probes (10:1): one across A–B for v_t, one across the load bank for v_R, both directly to the oscilloscope. Current probe (100 mV/A, zeroed) on the inductor wire with its arrow pointing from A toward the inductor, into the BNC splitter, which feeds the oscilloscope and the Fluke 39 current input. Fluke 39 voltage leads across the load bank. The interface-board probe inputs are not used.
5. E3631A: +15 V, 1 A limit, output on; check that LEDs D5, D9, D4 and the thermal-trip LED D8 are lit.

LaunchPad power is as in Lab 1: the +15 V supply must be on before you build and flash or connect External Mode, and turning it off resets the program to its built-in values. After any reset, reconnect External Mode, check that enableswitch selects disable and re-enter the command before continuing.

Software

1. Three programs are used: square_phase_shift_ti_launchpad for Part A (Experiments 1–2), spwm_bipolar_ti_launchpad for Experiments 3–4 and spwm_unipolar_ti_launchpad for Experiment 5. Build and flash each per PLECS Setup §4 and connect External Mode per §5. Leave enableswitch selecting disable. The tunable command is alpha (Part A) or m_i (Part B).
2. **Startup** (every experiment): enableswitch at disable → enter the starting command the experiment gives → ITECH output on at 55 V (the DC link charges through the current limit) → select enable → move the command to its test value.
3. **Changing programs** (before Experiments 3 and 5): follow Shutdown steps 1–2 (DC link below 10 V), flash the next program, then follow the startup sequence. No re-wiring is needed between experiments, except where an experiment says to move a probe.

STOP: TA to verify the circuit and instrument settings before the ITECH output is turned on.

Part A — Square-Wave and Phase-Shift Modulation

All experiments: v_d = 55 V, L = 5 mH, f_1 = 60 Hz. For each experiment, capture v_t, v_R and i_L over at least two fundamental periods (label each trace), record the Fluke 39 readings the questions ask for, and answer the questions in your report.

Experiment 1

Table 2: Experiment 1 parameters

Modulation	Program	α	R [Ω]
Square wave	square_phase_shift_ti_launchpad	0.5 (start), 0	12.5

Start with α = 0.5 (zero output), enable, then set α = 0: the square wave.

Questions:

- Explain the shape of i_L using the load time constant L/R, and compare it with your pre-lab sketch.
- Record $I_{L,1}$, $I_{L,3}$, $I_{L,5}$, $I_{L,7}$, the THD of i_L and the power delivered to R. Compare $I_{L,1}$, P_1 and $I_{L,3}$ with your pre-lab values and account for any deviation.

Experiment 2

Table 3: Experiment 2 parameters

Modulation	Program	α	R [Ω]
Phase shift	(as Experiment 1)	pre-lab value, 1/6	12.5

Continue from Experiment 1; no re-flash is needed. Set α to your question 2 value, then to 0.167 (1/6). The program applies α in steps of 1/120, so the α it uses may differ slightly from the value you enter.

Questions:

- At your pre-lab α: record $I_{L,1}$, $I_{L,3}$, $I_{L,5}$, $I_{L,7}$, the THD of i_L and the total power in R. Compare $I_{L,1}$, P_1 and $I_{L,3}$ with question 2, and compare the total power in R with P_1.
- At α = 1/6: record the same quantities. Compare the 3rd harmonic current with Experiment 1 and explain the result, including anything that remains. Tabulate $I_{L,1}$, $I_{L,3}$, $I_{L,5}$, $I_{L,7}$ and the THD of i_L at α = 0, 1/6 and your pre-lab value, and use these harmonic currents to explain how the THD varies with α.

Part B — Sinusoidal PWM

All experiments: v_d = 55 V, L = 5 mH, f_1 = 60 Hz, f_{sw} = 4.98 kHz. Capture each operating point in both the fundamental and the switching-detail views of Table 1. At m_i = 0.88 and 12.5 Ω in Experiments 3 and 5, also record $I_{L,1}$, the power in R, the THD of i_L, and the order and rms current of the largest low-order harmonic from the Fluke 39 for the Comparison at the end. In Experiments 3 and 5, also measure the frequency and peak-to-peak value of the i_L ripple at the positive peak of i_L in the switching-detail view.

Experiment 3

Table 4: Experiment 3 parameters

Modulation	Program	m_i	R [Ω]
Bipolar SPWM	spwm_bipolar_ti_launchpad	0.2 (start), 0.88	12.5

Questions:

- Compare the waveforms of v_t and v_R, and explain the difference.
- Using the oscilloscope FFT of i_L (Instrument Settings; Bench Instruments Guide §4), measure the frequency and rms value of the three largest components near f_{sw} and the two largest near $2f_{sw}$, and compare the largest with question 3(a).
- Using your oscilloscope FFT readings, estimate the THD of i_L including the components near f_{sw}, and compare it with the THD the Fluke 39 reports. Explain the difference.
- Compare the low-order harmonic currents on the Fluke 39 (up to the 31st) with what ideal SPWM predicts. If there are components the theory does not predict, propose at least two possible causes in the circuit or the modulator. After the lab, test the cause you consider most likely by changing SPWM_bipolar.plecs, and report the result.

Experiment 4

Table 5: Experiment 4 parameters

Modulation	Program	m_i	R [Ω]
Bipolar SPWM	(as Experiment 3)	0.88	12.5
Bipolar SPWM	(as Experiment 3)	0.44	12.5
Bipolar SPWM	(as Experiment 3)	0.88	50 (5 switches)

Continue from Experiment 3; no re-flash is needed. At each operating point, measure the power at the bridge output with the oscilloscope: define Math as channel 1 × channel 3 ($v_t \cdot i_L$) and read its average, and read the rms values of v_t and i_L, all over whole fundamental periods. Reach 50 Ω by opening switches one at a time.

Questions:

- Tabulate P, $V_{t,rms}$, $I_{L,rms}$, the power factor $P/(V_{t,rms}I_{L,rms})$ and the load's displacement factor $\cos \phi_1$ at the three operating points, and compare the m_i = 0.88, 12.5 Ω point with question 3(b).
- Explain how the bridge-output power factor compares with $\cos \phi_1$, and how it changes with m_i and with R.

Experiment 5

Table 6: Experiment 5 parameters

Modulation	Program	m_i	R [Ω]
Unipolar SPWM	spwm_unipolar_ti_launchpad	0.2 (start), 0.88	12.5

While the ITECH output is off for the program change (DC link below 10 V), move the v_R probe to measure the leg-A pole voltage v_{AN} (A to DC−); the Fluke 39 stays across R. Capture v_t, v_{AN} and i_L in both views of Table 1, triggering on i_L (rising, 0 A) in the fundamental view.

Questions:

- Compare v_t and the i_L ripple with Experiment 3: voltage levels, and the frequency and peak-to-peak value of the ripple at the i_L peak. From v_{AN}, at what frequency does each leg switch? Reconcile this with the ripple frequency of v_t and i_L.
- From the oscilloscope FFT of i_L, measure the frequencies and rms values of the largest components. Compare their frequencies with pre-lab question 4 and their rms values with the components at the same frequencies in Experiment 3. Compare the component at f_{sw} with Experiment 3.
- Measure the bridge-output power factor at $m_i = 0.88$ by the Experiment 4 method and compare it with bipolar SPWM. Explain the difference.
- Compare the Fluke 39's low-order harmonic currents with Experiment 3, and relate any difference to your explanation there.

Comparison

Questions:

- Tabulate, for the square wave (Experiment 1), phase shift at your pre-lab α (Experiment 2), and bipolar and unipolar SPWM at $m_i = 0.88$ (Experiments 3 and 5), all at 12.5 Ω: $I_{L,1}$, the power in R, the largest low-order harmonic current ($h \le 31$) and the THD of i_L from the Fluke 39, and, for the SPWM schemes, the largest component near f_{sw} or $2f_{sw}$ from the oscilloscope FFT. Rank the four schemes by the harmonic content of i_L and explain the ranking from how each one forms v_t. How many times does each switch turn on per fundamental period in each scheme, and what does that imply for switching losses? Weigh this against the harmonic ranking.

Shutdown

1. Select disable using enableswitch. Leave the load connected.
2. Ramp the ITECH voltage down to zero — the ramp discharges the DC link through the source. Watch the ITECH voltage reading, which is the DC-link voltage, and turn the output off only once it is below 10 V.
3. Turn off the E3631A, the current probe and the Fluke 39, disconnect your wiring, and return the lab equipment as instructed by the TA.

Lab Report

Submit one report per group on D2L by 10:00 AM one week after your lab session. The report is marked out of 10 using the rubric below, rather than experiment by experiment. Include the required captures and measurements, answer the questions directly, and use your results to support your explanations. Do not narrate the procedure.

Table 7: Lab report rubric (10 marks)

Criterion	Marks	Full-marks standard
Experimental evidence	3	Required captures and measurements are present and usable. Channels, scales, zero levels, probe factors, operating conditions and units are clear where applicable. Relevant measurement limitations are identified.
Analysis and comparison	3	Calculations are correct. Measurements are compared quantitatively with applicable theoretical predictions, with assumptions stated. Relevant trends across operating conditions are identified.
Physical interpretation	4	Answers explain the observed behaviour using switching states, current paths and circuit principles. Significant discrepancies are discussed using the available evidence, with uncertainty acknowledged.

Table 8: Quality scale, applied separately to each criterion

Level	Share of marks	Standard
Excellent	90–100%	Complete and technically correct; no substantive weaknesses.
Good	75–89%	Substantially complete and correct; minor weaknesses do not undermine the results or conclusions.
Adequate	60–74%	Demonstrates the essential requirements, but omissions or errors limit the analysis, interpretation or usefulness of the evidence.
Marginal	50–59%	Provides some relevant evidence or understanding, but substantial omissions or errors undermine the results or conclusions.
Inadequate	below 50%	Provides little usable evidence of meeting the criterion, or contains fundamental errors. No relevant work earns zero.